Jean-Pierre van Zyl, Ph.D. Candidate

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Employment History

Feb ’25 – Dec ’25	> **Stellenbosch University**. Part-time Junior Lecturer, MEng (Data Science) - Data Analytics, Applied Machine Learning, and Applied Deep Learning
Jul ’23 – Dec ’24	> **Stellenbosch University**. Teaching Assistant, MEng (Data Science) and BScHons (Computer Science)
Jul ’21 – Jul ’23	> **Stellenbosch University**. Tutor, Data Analytics and Applied Machine Learning
Mar ’21 – Aug ’21	> **Merlynn Intelligence Technologies**. Part-time developer
Jan ’21 – Feb ’21	> **Polymorph Systems**. Intern
Nov ’20 – Dec ’20	> **NMRQL Research**. Intern

Education

2023 – present	> **Ph.D. Computer Science**. Stellenbosch University.
2021 – 2022	> **M.Sc. Computer Science**. Stellenbosch University.
2020 – 2020	> **B.Sc.Hons. Computer Science**. Stellenbosch University.
2017 – 2019	> **B.Sc. Computer Science**. Stellenbosch University.
2012 – 2016	> **National Senior Certificate**. Fairmont High School.

Research Publications

[1] G.J. Steyn, **J. van Zyl**, and A.P. Engelbrecht, “Set-based particle swarm optimisation convergence,” Swarm Intelligence, vol. 20, no. 1, pp. 1–33, 2026.

[2] **J. van Zyl** and A.P. Engelbrecht, “Analysis of classification metric behaviour under class imbalance,” Egyptian Informatics Journal, vol. 31, p. 100 711, 2025.

[3] **J. van Zyl** and A.P. Engelbrecht, “Closed-form expressions for the normalizing constants of the mallows model and weighted mallows model on combinatorial domains,” Mathematics, vol. 13, no. 19, 2025.

[4] **J. van Zyl**, “Rule induction with swarm intelligence,” Magister Scientiae, Stellenbosch University, 2023.

[5] **J. van Zyl** and A.P. Engelbrecht, “Set-based particle swarm optimisation: A review,” Mathematics, vol. 11, no. 13, 2023.

[6] **J. van Zyl** and A.P. Engelbrecht, “Rule induction using set-based particle swarm optimisation,” in Congress on Evolutionary Computation, 2022, pp. 1–8.

[7] **J. van Zyl** and A.P. Engelbrecht, “Polynomial approximation using set-based particle swarm optimization,” in International Conference on Swarm Intelligence, Springer, 2021, pp. 210–222.

Miscellaneous Experience

2019 – 2022	>	**Student Leadership**. Served on the House Committee of student residence for four years as member, Vice-Primarius, and Primarius (head of house).
2021 – 2022	>	**Student Leadership**. Served on the executive committee of the Stellenbosch University Consulting Society.
2024 – present	>	**Volunteer**. Member of South African National Parks (SANParks) Honorary Rangers.

References

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